AI-DRIVEN MAINTENANCE REPORT

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1. Technical Sensor Data

Parameter	Measured Value
Air Temperature	298.1 °F
Process Temperature	308.6 °F
Rotational Speed	1200 RPM
Torque	75.2 Units
Tool Wear	40 %

2. AI Agent Analysis

Agent 1 (Initial Analysis):
Torque exceeds threshold, indicating high risk.

Agent 2 (Verification):
Torque exceeds 60 Nm, triggering high risk per rules.

3. Engineering Analysis & Recommendations

Summary:
REQUIRED MAINTENANCE

Analysis:
The sensor data indicates that the system is experiencing high torque, exceeding the safety threshold of 60 Nm. This is a critical condition that requires immediate attention to prevent potential damage or failure.

Validation:
The validation checks confirm that the torque value (75.2) exceeds the threshold (60), indicating a high risk situation. Additionally, the tool wear level (40) is above the acceptable range, further emphasizing the need for urgent maintenance.

Recommendation:
1. IMMEDIATELY REDUCE TORQUE LOAD: Reduce the rotational speed or adjust the system's load to minimize the torque and prevent further damage.
2. PERFORM URGENT TOOL INSPECTION AND REPLACEMENT: Inspect the tool for signs of wear and tear, and replace it if necessary to ensure safe operation.
3. SCHEDULE FURTHER MAINTENANCE: Schedule a comprehensive maintenance check to identify and address any underlying issues contributing to the high torque condition.

4. Conclusion

Final Decision: REQUIRED MAINTENANCE
Decision Source: RULE-BASED SYSTEM TRIGGERED